

Assignment 1 Instructions

PSYC417

Assignment 1

This assignment is going to test your skills in data types, data structures, conditionals, and loops. You may get to the answer **using any of the techniques we specify in class, unless the question requests otherwise. If so, you must use that technique.** Do not use techniques or code we didn't learn in class (e.g., from a package we haven't used or a function we haven't covered). This will result in a modest deduction.

The data for this assignment are a data frame consisting of students enrolling at UMD in the Fall, their intended major, SAT score, and whether they were in honors in college. You can use the below code to create the data frame (make sure the code isn't wonky after pasting it into R and fix it if necessary).

```
hs_students <- data.frame(  
  stud_ID = c(1:10),  
  major = c('PSYC', "PSYC", "NEUR", "CRIM", "NEUR", "GVPT", "GVPT", "JOUR", "PSYC", "ECON"),  
  SAT = c(1200, 1300, 1400, 1100, 1560, 1600, 1250, 1400, 1410, 1500),  
  honors = c("YES", "NO", "NO", "YES", "YES", "UNK", "YES", "NO", "UNK", "UNK")  
)
```

Questions

Write each question/number into the Quarto document as text and place any code or text to answer the question on the lines below it. In short, it should look something like below:

Question 1 - Text from question 1. (MAKE THIS ONE OF THE HEADER STYLES SO IT'S EASIER TO READ.)

Code chunk, if applicable.

Text answering question, if applicable.

Question 2 - Text from question 2. (MAKE THIS ONE OF THE HEADER STYLES SO IT'S EASIER TO READ.)

.... and so on.

Description

Question 0 (5pts total)

Create a Quarto document titled "Assignment 1" with you listed as the author (1pt). **Make sure the document type is .docx (1pt).** When uploading the assignment to ELMS, ensure you upload both the .qmd and .docx files (3pts). (No fixing this after the deadline if you forget.)

Question 1 (30pts total)

Q1.a (5pts total)

How many columns are there in the data frame? Use two different commands to tell you that.

Q1.b (5pts total)

Create a NEW data frame called "SSC1" and (using brackets on the original data frame) have it only contain the PSYC and NEUR students. Create a SECOND data frame called "SSC2" and (using the function we learned on the original data frame) have it contain the PSYC and NEUR students. (Basically, use the two techniques we learned to subset the PSYC and NEUR students.)

Q1.c (9pts total)

Create an object called all_mean and make it a vector equal to the SAT scores from hs_students (3pts). Create an object called psyc_mean and make it a vector equal to the SAT scores from SSC1 (3pts). (Do not manually enter the data for either.) With a logical operator, compare the averages using the mean() function in R (3pts).

Q1.d (11pts total)

Create a new column in “hs_students” called “new_honors” that consists of logical values. Have it be TRUE if honors was a YES and FALSE otherwise. (Do not manually enter the data here.)

DO NOT USE ANY PRE-EXISTING COMMANDS LIKE MEAN(), MIN(), OR SD() FROM HERE ON OUT. You will use these to check your answers, but you must hard-code everything.

Question 2 (25pts total)

Find the minimum value of the all_mean vector using a for loop. Hint: You will need to create an object outside of the loop that acts as a baseline MINIMUM value. (Think the largest value in R from the first set of slides.)

Question 3 (25pts total)

Compute the average value of the PSYC_mean vector using a for loop. Hint: You will need to create objects outside of the loop that allows you to tally the sum and number of observations. You can use the length() command when defining the loop, but you cannot use it in the denominator of the mean formula.

Question 4 (15pts total)

Create an object called “SAT_scores” and set it equal to the SAT scores from the hs_students data frame. USING A LOOP AND CONDITIONAL STATEMENTS, cycle through the SAT_scores vector and print whether a score is “OKAY” if it less than 1200, “GOOD” if it is between 1200 and 1300 (inclusive), and “GREAT” if it is above 1300. (Remember the guidance about else vs. else if statements.)

The End

Render the document and upload the .qmd and .docx files to ELMS.

Thank you! :)